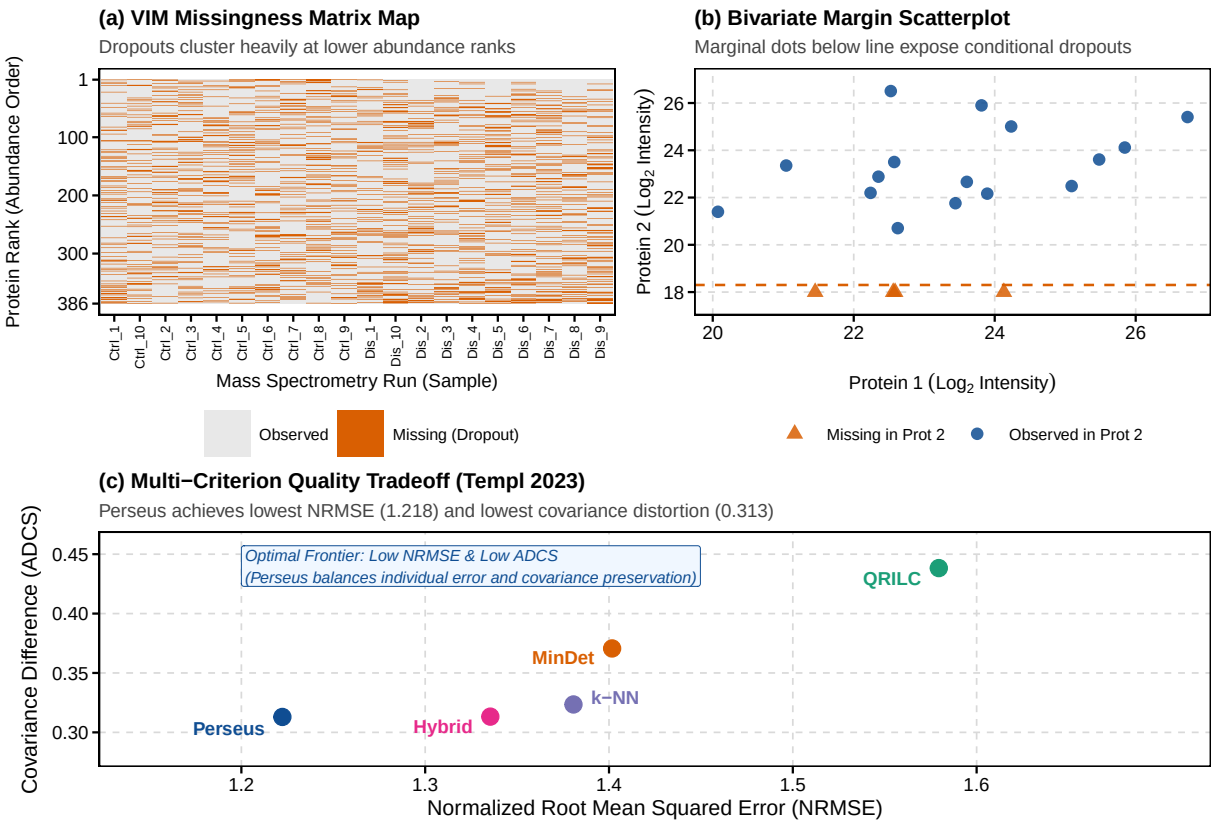
(a) VIM Missingness Matrix Map
Dropouts cluster heavily at lower abundance ranks
Protein Rank (Abundance Order)
1
100
200
300
386
Ctrl_1
Ctrl_10
Ctrl_2
Ctrl_3
Ctrl_4
Ctrl_5
Ctrl_6
Ctrl_7
Ctrl_8
Ctrl_9
Dis_1
Dis_10
Dis_2
Dis_3
Dis_4
Dis_5
Dis_6
Dis_7
Dis_8
Dis_9
Mass Spectrometry Run (Sample)
Observed
Missing (Dropout)
(b) Bivariate Margin Scatterplot
Marginal dots below line expose conditional dropouts
Protein 2 (Log2 Intensity)
18
20
22
24
26
Protein 1 (Log2 Intensity)
Missing in Prot 2
Observed in Prot 2
(c) Multi−Criterion Quality Tradeoff (Templ 2023)
Perseus achieves lowest NRMSE (1.218) and lowest covariance distortion (0.313)
Optimal Frontier: Low NRMSE & Low ADCS
(Perseus balances individual error and covariance preservation)
Covariance Difference (ADCS)
0.30
0.35
0.40
0.45
QRILC
MinDet
k−NN
Hybrid
Perseus
1.2
1.3
1.4
1.5
1.6
Normalized Root Mean Squared Error (NRMSE)